

Phase 3 — Option C

Final Implementation Report

CareerBuddies Founder Finance · expense-linked reimbursement

Branch `claude/compassionate-bardeen-9zxrjv` · code commit `8c732ac`

Final status: **PHASE 3: APPROVED** · Phase 4 not started

1 Project

CareerBuddies Founder Finance — private web app for the three CareerBuddies founders (investment, expenses, settlements). Monorepo `CB-Founder-Ledger/` in `d4deepak-2005/career-buddies-website`, branch `claude/compassionate-bardeen-9zxrjv`. Stack: React 19 · TypeScript · Vite · Tailwind — Node 22 · Express · Mongoose · Zod — MongoDB 7 — Docker Compose.

2 Phase

Phase 3 — Calculation engine, finalised with **Option C (expense-linked reimbursement)**. Phases 1 and 2 are complete and unchanged except for the documented Phase 2 amendment. **Phase 4 has not been started.**

3 Final policy

- A reimbursement **must** reference exactly one approved business expense (`reimbursesTransactionId`); the field is forbidden on every other type.
- The linked expense must have exactly one payer; the reimbursed founder must be that payer.
- The reimbursed portion is **business-borne**. Founder-funded = expense – Σ active/approved linked reimbursements, split by the expense's stored split with the existing largest-remainder rounding.
- Net position = Paid – Fair share (*inferred from the Product Plan example*). Σ net positions = 0 exactly.
- Over-reimbursement is rejected, including concurrent attempts. An expense with active linked reimbursements cannot be voided. Voiding a reimbursement restores the founder-funded amount. Nothing is deleted.
- No fake founder, no capital pool, no attribution rule (IA-5 removed), no `PASS_WITH_EXTERNAL` status.

4 Decision

The product owner chose Option C after the feasibility review (docs/PHASE-3-OPTION-C-FEASIBILITY.md). Labelled **implementation assumption — product-owner decision**: the Product Plan PDF does not say how a reimbursement affects founder balances. Decision 1 in docs/PHASE-3-OPEN-DECISIONS.md is marked resolved; Decisions 2–5 (founder-to-founder reimbursement, capital pool, loan repayment, refund linking) remain open and unchanged.

5 Architecture

The backend calculation engine (server/src/domain/calculationEngine.ts) is the single source of truth; the frontend only formats and displays. Write-time safety lives in modules/transactions/reimbursements.ts ; the engine independently recomputes everything from the linked records and never trusts the stored counter.

Layer	Responsibility
Engine (pure)	Validates, links reimbursements to expenses in (date, number, id) order, applies cap, scales stored splits, builds positions, reconciliation, settlement recommendations
reimbursements.ts	Target checks, atomic capacity reservation/release, linked lookup, picker query
transactions.service	Create/edit/void hooks keep the reservation exact (rollback on failure)
financials.service	Loads data, calls the engine, shapes read-only responses
Client	Expense Picker, detail links, "reimbursed by the business" wording; no money arithmetic

6 Data model

Field	On	Notes
reimbursesTransactionId	reimbursement	ObjectId → Transaction; required for reimbursements, forbidden otherwise; cannot be removed once set; indexed {reimbursesTransactionId:1, status:1} (sparse)
reimbursedMinor	business_expense	Internal reservation counter (integer minor units, default 0, ≥ 0); never client-writable; the engine does not read it
Money	all	Integer minor units (₹30.00 = 3,000); no floating point

7 API

Endpoint / change	Behaviour
POST /api/transactions , PATCH /api/transactions/:id	Accept reimbursesTransactionId for reimbursements; strict Zod schema; operators/dotted keys rejected
GET /api/transactions/reimbursable-expenses?paidByFounderId=... [&forReimbursementId=...]	Approved expenses of that payer with remaining capacity (picker); strict query
GET /api/transactions/:id	Adds reimbursesTransaction , reimbursedMinor , remainingReimbursableMinor , linkedReimbursements[]

Endpoint / change	Behaviour
POST /api/transactions/:id/void	409 HAS_LINKED_REIMBURSEMENTS for an expense with active reimbursements
Errors	400 REIMBURSEMENT_EXCEEDS_EXPENSE (with remainingMinor), UNKNOWN_TARGET, TARGET_NOT_EXPENSE, TARGET_NOT_APPROVED, PAYER_MISMATCH
Financial endpoints	Unchanged paths; businessBorneMinor replaces externalMinor ; read-only, empty query only

8 Calculation engine

Per valid approved expense E (amount X , payer P): $R = \sum \text{valid linked reimbursements}$, $\text{founderFunded} = X - R$, $\text{shares} = \text{stored allocations scaled to founderFunded (largest remainder)}$; $\text{paid}(P) += X - R$. Exclusions are named warnings: `REIMBURSEMENT_NOT_LINKED`, `_TARGET_MISSING`, `_TARGET_INVALID`, `_TARGET_NOT_OFFICIAL`, `_PAYER_MISMATCH`, `_EXCEEDS_EXPENSE`. Reconciliation checks: `NET_POSITIONS_ZERO_SUM`, `SETTLEMENTS_ZERO_SUM`, `RECEIVABLE_EQUALS_PAYABLE`, `FAIR_SHARE_RECONCILES`, `PAID_RECONCILES`, `REIMBURSEMENTS_WITHIN_EXPENSES`, `RECOMMENDATIONS_CLEAR_BALANCES`.

₹3,000 expense paid by A, equal 3-way	Paid A/B/C	Fair share A/B/C	Net A/B/C
No reimbursement	3,000 / 0 / 0	1,000 each	+2,000 / -1,000 / -1,000
₹1,000 reimbursed	2,000 / 0 / 0	666.67 / 666.67 / 666.66	+1,333.33 / -666.67 / -666.66
Fully reimbursed	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0
₹1,000 + ₹500	1,500 / 0 / 0	500 each	+1,000 / -500 / -500

Concurrency: capacity is reserved by one conditional `$expr` update on the expense before the reimbursement is inserted; released on rollback, void, edit-down/away/move. The expense void write is conditional on `reimbursedMinor ∈ {0, null}`. `npm --prefix server run reconcile:reimbursements` reports counter drift.

9 UI

Add/Edit Transaction: choosing *Reimbursement* shows an **Expense Picker** (approved expenses of the founder being reimbursed, each with the amount still reimbursable; explains when none are left). The client sends only the expense id, never a financial value. Server errors (over-reimbursement, payer mismatch) are displayed verbatim. Detail pages link reimbursement ↔ expense and show "reimbursed / still reimbursable / founders share"; a blocked expense void shows the server explanation. Founder, ledger and settlement pages say "reimbursed by the business".

Mobile — picker, missing-expense message

Tablet — over-reimbursement rejected

Desktop — positions after ₹1,000 of ₹3,000 reimbursed

10 Validation

- Link required for reimbursements; forbidden for all other types (create and edit).
- Target exists, is a business expense, is approved, has a payer, payer equals the reimbursed founder.
- Σ active reimbursements $\leq$ expense amount (draft, pending and approved hold capacity); checked atomically.
- Zod `.strict()` schemas, ObjectId validation, integer positive amounts.

11 Security

- No financial value is accepted from the client: the reservation counter is not in any request schema; calculation endpoints are read-only and reject any query string.
- NoSQL operator injection in the link field (e.g. `{"$ne": null}`) is rejected (tested over HTTP and in the API suite).
- Server-built operators use Mongoose `trusted()` under `sanitizeFilter`.
- Unchanged: httpOnly SameSite=Strict cookies, rate limits, RBAC, private receipt storage, no hard deletes (history in append-only revisions). MongoDB is never published to the host.

12 Test strategy

Hand-computed unit tests for the engine; hundreds of random scenarios over 4 founders for invariants (Σ net = 0, business-borne = Σ valid reimbursements, no money created or lost, excluded statuses change nothing, order independence); API tests on real MongoDB 7; client component tests with a stubbed API; Docker smoke + business-flow E2E; Playwright browser flow at three viewports with API-equals-display checks; responsive stress. No existing test was deleted or weakened; tests that encoded the superseded IA-5/external behaviour were rewritten to Option C figures.

13 Server tests (exact)

279 passed / 279, 17 files, run against a real MongoDB 7 container.

File	Tests
accountingReview	37
auth	12
calcInvariants	8
calculationEngine	50
config	5
database	5
financials	19
password	3
rateLimit	1
rbac	22
receipts	13
reimbursements (new)	25
settlementAlgorithm	7
splits	15
transactionRules	7
transactions	40
validation	10

Server `tsc --noEmit` : clean.

14 Client tests (exact)

53 passed / 53, 5 files (App 9, api 4, money 4, financials 15, transactions 21). Client `tsc` clean; production build succeeds. New: expense picker (requires a choice, lists only the payer's reimbursable expenses, empty state, sends only the link id, shows server over-reimbursement error), detail links and reimbursed summary, blocked-void message, business-borne wording.

15 API tests (exact)

The 25-test `reimbursements` suite plus the updated `financials` (19) and `transactions` (40) suites — all through HTTP on MongoDB. They cover all 30 required cases:

Required case	Covered by
linked; mandatory link; non-reimbursement cannot carry link; reimbursement cannot exist without link	reimbursements 1–3, transactions "4. Reimbursement"

Required case	Covered by
wrong target type; invalid/deleted target; unapproved target; payer mismatch	reimbursements 4, 21
₹1,000 of ₹3,000; full ₹3,000; multiple partial; cumulative cap; over-reimbursement rejected	reimbursements 5, 6, 7, 9
reimbursement void restores; expense void blocked; expense void after reimbursement void	reimbursements 13–16
zero-sum; largest-remainder rounding; different splits per expense	reimbursements 10; calcInvariants A1–A2; accountingReview
concurrent protection (5 parallel; 2 parallel; void-vs-create race)	reimbursements 20
settlement after partial/full reimbursement	reimbursements 22; accountingReview 22–24
refund / loan / contribution regression; Other exclusion; voided/pending/rejected exclusion	reimbursements 11–12; accountingReview 15–17

16 MongoDB verification (exact)

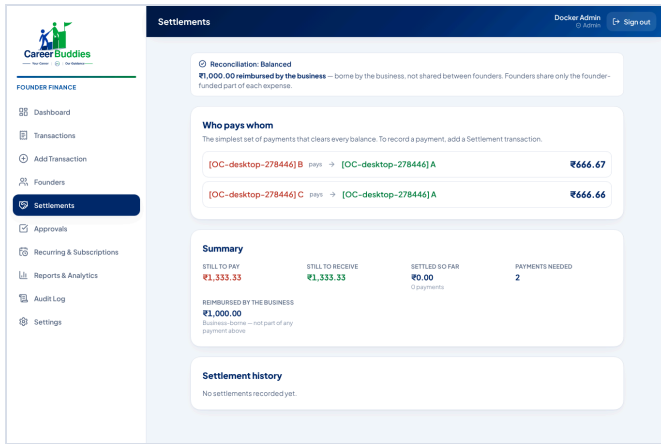
Inspected directly in the Docker MongoDB after the full run (`mongosh` through the compose network): reimbursement documents without a link **0**; non-reimbursements carrying a link **0**; expenses with `reimbursedMinor > amountMinor` **0**; index `reimbursesTransactionId_1_status_1` present; every test transaction in status `voided` (none deleted; revision history intact); `reconcile:reimbursements` : *"No drift: every expense counter matches its active reimbursements."* The normal dev DB holds no test financial data; test databases and the Docker volume were removed.

17 Docker verification (exact)

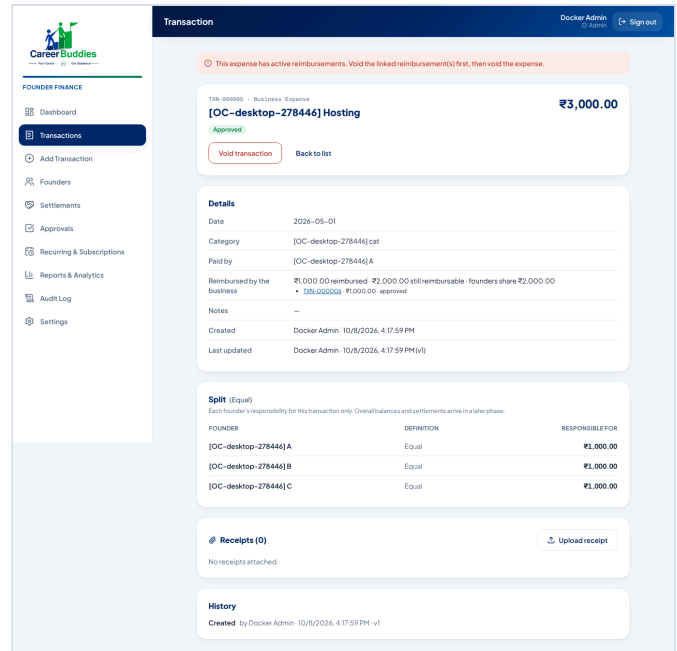
Images built; Compose stack **client=healthy, mongo=healthy, server=healthy**; `/api/health` reports the database connected. **Smoke: 61 passed / 0 failed** (includes 10 new reimbursement-validation checks: missing link, unapproved/unknown/malformed target, operator injection, expense cannot carry link, picker authentication and validation). **Reimbursement E2E over HTTP: 22 passed / 0 failed** (create, picker, payer mismatch, partial reimbursement, exact figures, recommendations, over-reimbursement rejection with remaining amount, void guard, void restore, expense void). Approval uses a direct DB write inside the compose network because Phase 5 owns approval.

18 Browser testing (exact)

Playwright + Chromium against the Docker stack: **99 checks passed / 0 failed** (33 per viewport × 3), run twice in a row with identical results. Flow: login, expense fixture approval, Expense Picker (payer-first message, option shows remaining amount), submit without expense (message), reimbursement creation, detail link back to the expense, capacity reserved (DB), founder positions (A paid 2,000.00 / fair 666.67 / net +1,333.33; B -666.67; C -666.66), API values equal displayed values, Σ net = 0, settlement recommendations B→A 666.67 and C→A 666.66, "no approved expenses left" explanation, over-reimbursement (₹2,500 > ₹2,000 left) rejected with nothing reserved, expense void blocked and still approved, reimbursement void (record kept, capacity released, A back to paid 3,000.00 / fair 1,000.00 / net +2,000.00), expense void succeeds, no console errors.



Desktop — recommended payments



Desktop — expense void blocked

19 Responsive testing (exact)

Desktop 1366×900, tablet 820×1180, mobile 390×844: no horizontal overflow on the reimbursement form, reimbursement detail, expense detail, founders and settlements pages (18 checks inside the 99: 6 pages × 3 viewports). Stress: **300 page loads** (founders, settlements, transactions, add transaction, settings × 15 × 4 viewports incl. 320×640): **0 overflows**. **This testing found and fixed a real defect:** with a long founder name the Founders page overflowed at 390px (434px wide) because the single-column grid track used min-content width; fixed with `grid-cols-1` and `min-w-0` on the card, verified at 390px (scrollWidth 390).

20 Public testing URL

Not available — stated limitation. This build environment is a sandbox container with outbound access only through a policy proxy (the connectivity probe to a tunnel service returned HTTP 403/000) and no inbound exposure or tunnel tooling. No public deployment was attempted and **no public URL is claimed**. Exposing MongoDB or real credentials publicly was not considered. Everything above was tested on the local Docker stack:

```
cd CB-Founder-Ledger
cp .env.example .env # set JWT_ACCESS_SECRET, MONGO_ROOT_PASSWORD, SEED_ADMIN_EMAIL,
SEED_ADMIN_PASSWORD; for plain HTTP: NODE_ENV=development, COOKIE_SECURE=false
docker compose --env-file .env -f docker/docker-compose.yml up --build -d
docker compose --env-file .env -f docker/docker-compose.yml exec server node
dist/scripts/seedAdmin.js
# open http://localhost:8080 (health: http://localhost:8080/api/health)
```

21 Test credentials

Safe, disposable, local-only credentials used for this verification (the stack and its database volume were destroyed afterwards): admin `admin@careerbuddies.test` / Docker-verify-passphrase-9. They work only on a stack seeded with those values. **No production credentials, secrets, API keys or database credentials appear in this report or the repository** (`.env` is git-ignored).

22 Files changed

36 files, +1,414 / -342 relative to the pre-implementation base (a202fb5), none outside CB-Founder-Ledger/ .

Area	Files
Server domain/model/API	domain/calculationEngine.ts (rewritten), domain/transactionRules.ts, models/Transaction.ts, modules/transactions/{reimbursements.ts (new), transactions.service.ts, transactions.routes.ts, transactions.schemas.ts}, modules/financials/financials.service.ts, scripts/reconcileReimbursements.ts (new), package.json
Server tests	reimbursements.test.ts (new), accountingReview, calcInvariants, calculationEngine, financials, transactions tests, calcFixtures
Client	lib/types.ts, features/transactions/{TransactionForm, TransactionDetailPage}.tsx, features/financials/{parts, FoundersPage, FounderLedgerPage, SettlementsPage}.tsx, tests, test/utlis.tsx
Verification	tests/smoke-docker.sh, tests/docker-reimbursement-e2e.sh (new), tests/browser/optionc-browser.mjs (new)
Docs	README.md, docs/{ARCHITECTURE, PHASE-2, PHASE-3-CALCULATION-SPEC, PHASE-3-OPEN-DECISIONS, PHASE-3-OPTION-C-FEASIBILITY}.md, this report

23 Documentation updated

README, ARCHITECTURE, PHASE-2 (amendment), PHASE-3-CALCULATION-SPEC (rule register rewritten: IA-4 revised, IA-5 removed, IA-5b/5c added, formulas, reconciliation, API, tests), PHASE-3-OPEN-DECISIONS (Decision 1 resolved), PHASE-3-OPTION-C-FEASIBILITY (marked approved + implementation section). Each rule is labelled PDF requirement / inferred from PDF example / implementation assumption / Phase 3 limitation.

24 Assumptions

- **Implementation assumption (product-owner decision):** reimbursement is business-funded, expense-linked, and the reimbursed part is not shared between founders.
- **Implementation assumption:** draft, pending and approved reimbursements hold capacity; voided and rejected do not.
- **Inferred from the Product Plan example:** Net Position = Paid – Fair Share; positive = receive, negative = pay.
- **PDF requirement:** only approved transactions are official; approved records are never silently deleted.

25 Limitations

- **Phase 3 limitation:** nothing can become *approved* through the app until Phase 5, so a real deployment shows zero and reimbursement can only be exercised end-to-end with fixture approvals (direct DB write). Authorization was not weakened.
- One payer per expense; multi-payer expenses cannot be reimbursed.
- No loan repayment, refund linking, capital pool, founder-to-founder reimbursement model.
- Settlement paths are not guaranteed minimal in number.

26 Known issues

- No public URL (section 20).
- The auth rate limiter is in-memory and counts logins per IP (10 / 15 min); repeated automated browser runs needed a server restart between runs and the browser suite logs in once and reuses the session.
- The pre-Option-C browser scripts (`browser*.mjs` , kept outside the repository) encoded IA-5 figures and were superseded by `tests/browser/optionc-browser.mjs` ; they were not rerun.
- The reservation counter can drift only through manual database edits; detection is provided (`reconcile:reimbursements`), automatic repair is not.
- No other defects known.

27 Git

Branch `claude/compassionate-bardeen-9zxrjv` . Implementation commits: `f9dd526` (server), `c273f3f` (client), `8c732ac` (docs, Docker/browser verification, overflow fix). The commit adding this PDF follows immediately (a commit cannot contain its own hash); its hash is reported in the final message. Working tree clean after push.

28 Final status

PHASE 3: APPROVED

All required verification passed: server 279/279, client 53/53, typecheck and build clean, Docker smoke 61/61, Docker E2E 22/22, browser 99/99, responsive stress 300 loads / 0 overflows, MongoDB inspected directly. The one outstanding item is the unavailable public URL (environment limitation, not a product defect).

29 Phase 4 status

Phase 4 has not been started. No dashboard KPIs or charts, approval workflow (5), recurring expenses (6), reports (7), hardening (8) or launch (9) work was done.